

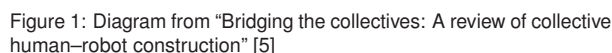
Isla Xi Han*

ABSTRACT

Index Terms: Human-centered computing—Interaction paradigms; Human-centered computing—Interaction paradigms—Virtual reality; Human-centered computing—HCI theory, concepts and models; Human-centered computing—Interaction design;

Over the past few decades, robotic technology has been adopted in the architectural field mainly as a tool to automate the fabrication process and to materialize complex parametric designs. However, little has been done to actively involve robots in the building-design process, which has the potential to 1) free humans from less important design decisions such as the exact layout of tiles, 2) inspire more creative formal languages from humans through human-robot collaboration, and 3) enhance the structural soundness and performance through real-time sensing and adapting. In addition, since most design works are finalized before the fabrication stage, how the interactions and collective behaviors that emerge from human-robot heterogeneous teams through the fabrication process can influence design decisions is yet to be explored.

*e-mail: xihan@princeton.edu



2 PRIOR PH.D. RESEARCH

My Ph.D. work so far has been focused on setting the theoretical and research foundation for Collective Human-Robot Collaboration (CHRC) in an improvisational setting. In the review paper published in the International Journal of Architectural Computing in 2021 [5], I defined the research gap between collective robotic construction and human-robot interaction in the building sector. The diagram in this review paper (Fig. 1) has identified “Virtual, Augmented, and Mixed Reality” as an essential platform of communication and collaboration between human and robotic collectives, thus central to the field of CHRC.

979-8-3503-4839-2/23/\$31.00 ©2023 IEEE
DOI 10.1109/VRW58643.2023.00327



Figure 2: Improv-Structure, in the middle of the building [7]

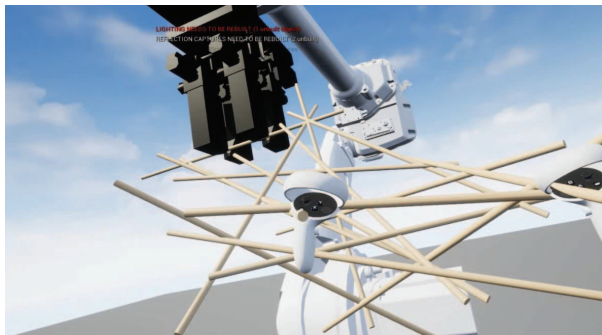


Figure 3: Basic setup of Improv-Structure 2.0 in virtual reality

structure to feel the space and make further decisions based on the spatial quality of the built part. To this point, my Ph.D. work has set up the foundational concept of CHRC and used prototype examples to demonstrate how the idea of improvisation can be brought into a multi-human-robot construction setting to facilitate the design-fabrication process. The last session of paper [7] points towards the possibility of transferring the Improv-Structure setting into virtual or mixed reality settings to explore how HRI for improvisational construction in virtual environments can help reduce the costs and risks that would have occurred in a physical construction yet remains the benefit of inspiring novel designs.

My most recent work, Improv-Structure 2.0, has started to transfer the Improv-Structure setting, which includes two ABB6400 robotic arms, bamboo rods, and human participants, into virtual reality using Unreal Engine and the Oculus Quest 2 VR headset. The current state of the VR environment encompasses player controllers that generate an infinite amount of bamboo rods to be placed at the designated 3D location and robotic arms with inverse kinematics (Fig. 3). Improv-Structure 2.0 is still under development.

3 PLANNED FUTURE WORK

The immediate next steps for Improv-structure 2.0 in VR are three-fold.

1. While VR makes it easier to collect the spatial information of the built structure compared to LiDAR scanning in physical reality, it is yet to be defined what spatial data is the most essential for the robots to learn human designer's building styles using a potential AI setup.
2. Explore how experiments between humans and robotics arms in VR can be transferred into useful knowledge in real-world constructions.

3. Set up tests to evaluate how creating with robotics arms in VR can influence how humans design.

In the longer term, I am interested in using the findings in VR design experiences to assist a smooth and intuitive blend between virtual and physical realities via mixed reality (MR). This will include topics such as creating meaningful interactions with the physical materials assisted by virtual spatial information (e.g., using Passthrough [2]).

In the Doctoral Consortium at IEEE VR 2023, I would like to discuss the following research questions further with the cohort:

- How to approach the lack of actuality, touch, and weight of building material for human designers in a virtual reality experience?
- How to carry the knowledge learned from virtual human-robot collaboration experiences into physical reality, given the misalignment between simulation and the physical world?
- How to optimize the 3DUI in a Collective Human–Robot Construction (CHRC) setting in virtual reality to inspire productive improvisation sessions?

ACKNOWLEDGMENTS

I would like to express my sincere gratitude to my advisors Prof. Stefana Parascho, Prof. Forrest Meggers, and Prof. Naomi Leonard, for their continuous support for my Ph.D. study and research.

I would also like to thank MoVA (Museum of Virtual Arts) for the equipment support of the Meta Quest 2 VR Headset.

REFERENCES

- [1] K. H. Ahlers, A. Kramer, D. E. Breen, P.-Y. Chevalier, C. Crampton, E. Rose, M. Tuceryan, R. T. Whitaker, and D. Greer. Distributed augmented reality for collaborative design applications. In *Computer Graphics Forum*, vol. 14, pp. 3–14. Wiley Online Library, 1995.
- [2] M. Banquero, G. Valdeolivas, S. Trincado, N. García, and M.-C. Juan. Passthrough mixed reality with oculus quest 2: A case study on learning piano. *IEEE MultiMedia*, 2023.
- [3] F. P. Brooks Jr. Walkthrough—a dynamic graphics system for simulating virtual buildings. In *Proceedings of the 1986 workshop on Interactive 3D graphics*, pp. 9–21, 1987.
- [4] J. Du, Z. Zou, Y. Shi, and D. Zhao. Zero latency: Real-time synchronization of bim data in virtual reality for collaborative decision-making. *Automation in Construction*, 85:51–64, 2018.
- [5] I. X. Han, F. Meggers, and S. Parascho. Bridging the collectives: A review of collective human–robot construction. *International Journal of Architectural Computing*, 19(4):512–531, 2021.
- [6] I. X. Han and S. Parascho. Block play with a robotic arm. In *SHERobots: Tool, Toy, Companion [exhibition catalogue]*, pp. 70–73. Sydney, Australia, 2022.
- [7] I. X. Han and S. Parascho. Improv-structure: Exploring improvisation in collective human-robot construction. In *International Conference on Trends on Construction in the Post-Digital Era*, pp. 233–243. Springer, 2023.
- [8] V. G. Kini and S. Sunil. Design and realization of sustainable rural housing using immersive virtual reality platform. In *25th ACM Symposium on Virtual Reality Software and Technology*, pp. 1–2, 2019.
- [9] D. Mitterberger, L. Atanasova, K. Dörfler, F. Gramazio, and M. Kohler. Tie a knot: human–robot cooperative workflow for assembling wooden structures using rope joints. *Construction Robotics*, pp. 1–16, 2022.
- [10] D. Mitterberger, K. Dörfler, T. Sandy, F. Salveridou, M. Hutter, F. Gramazio, and M. Kohler. Augmented bricklaying. *Construction Robotics*, 4(3):151–161, 2020.
- [11] M. J. Neil. Architectural virtual reality applications. *ACM SIGGRAPH Computer Graphics*, 30(4):53–54, 1996.
- [12] A. Webster, S. Feiner, B. MacIntyre, W. Massie, and T. Krueger. Augmented reality in architectural construction, inspection and renovation. In *Proc. ASCE Third Congress on Computing in Civil Engineering*, vol. 1, p. 996. Citeseer, 1996.